Make Robots Plan Faster: Evaluating Action Sampling Efficiency using Generative Models

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Abstract

To better perform everyday chores, such as organizing a bookshelf, **home robots** plan both *what* to do and *how* to do. Solving this challenge, known as **Task and Motion Planning (TAMP)**, combines a symbolic planner for high-level actions (e.g., *pick up the book*) with an action sampler that generates continuous motion parameters (e.g., how far to extend the arm). Recent work uses **Generative Models** as samplers, allowing robots to learn distributions of possible actions. However, poor samples can cause action failures, increasing planning time and reducing efficiency. This work investigates the **trade-off between sampling quality, the number of samples used, and their impact at task planning time and overall planning success** with learned samplers. These results explore how Generative Models can be integrated into real-world robots.

Background

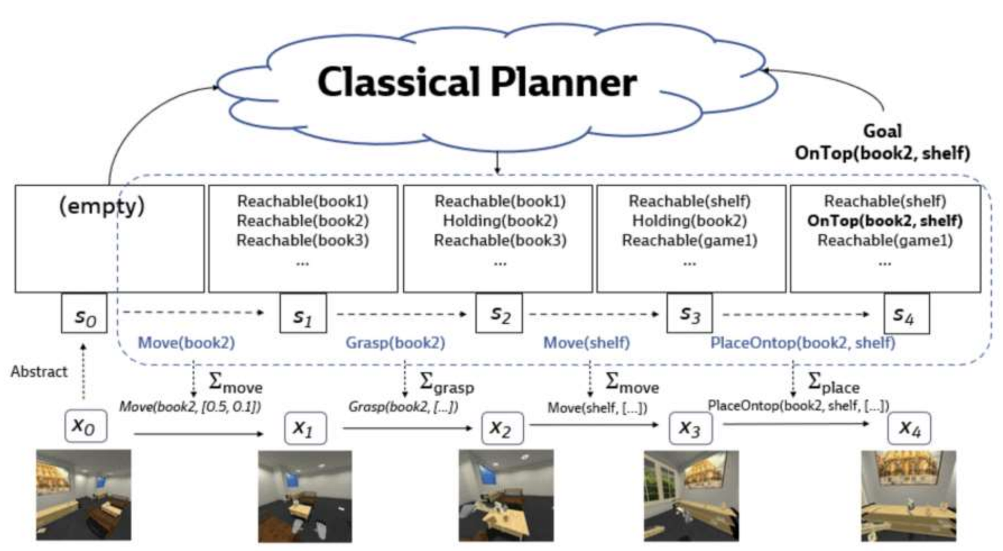


Figure 1: Bilevel Planners combine three components: (1) planner to determine **a sequence of actions** to perform, (2) motion planner to select specific actions, and (3) an action sampler to provide **parameters** for execution (Kumar et. al, 2023)

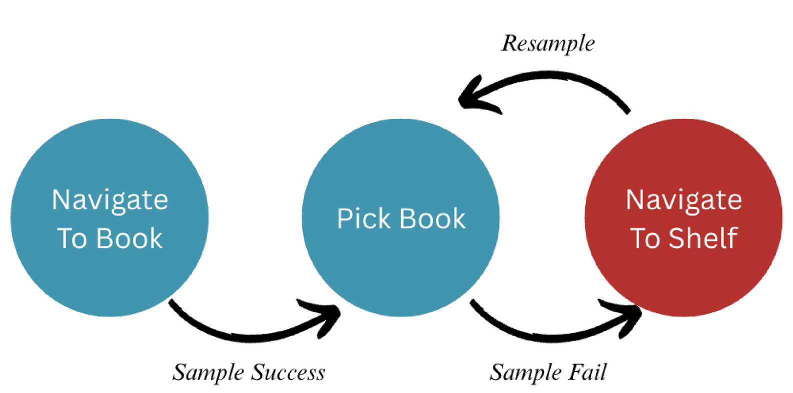


Figure 2: If action sampler produced successful parameters, the planner goes to the next state.

If the samples were incorrect, the algorithm **backtracks** to the previous state(s) and **resamples** until it fails or reaches the goal.

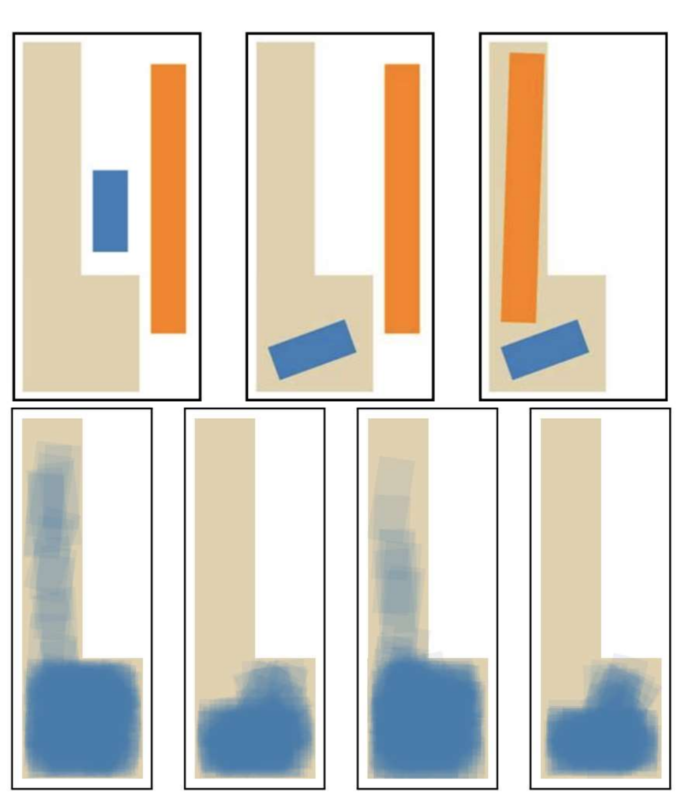


Figure 3: Action sampler learns to move two sticks into the space. Bottom blue shadows show the placements of blue stick.

"Valid" means possible locations where blue stick fits in, but "Success" shows locations of the blue stick where the orange stick also fits in. (Mendez et.al, 2023)

Models

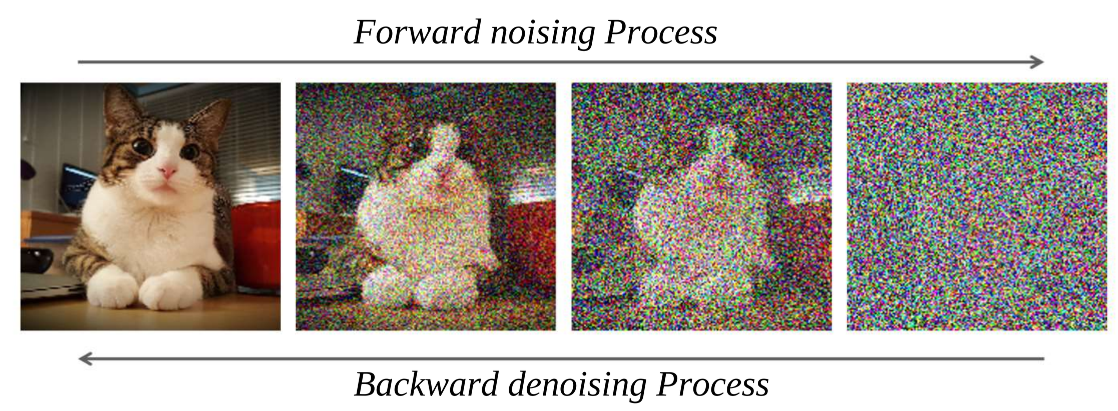


Figure 8: Diffusion models (DDPM) generate images. The forward process transforms an image to noise and the denoising process removes noise using neural networks **step by step** to "recover" a new image.

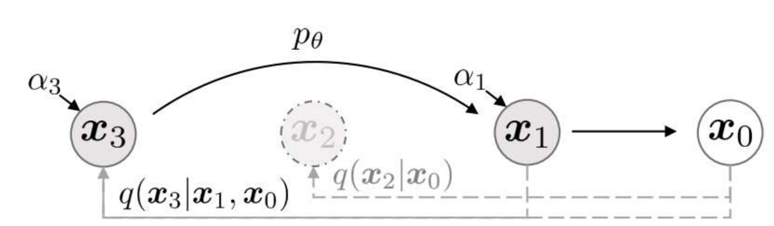


Figure 9: To speed up the sampling process, DDIM can **skip steps** during the sampling process while retaining similar level of performance. (Song et.al, 2021)

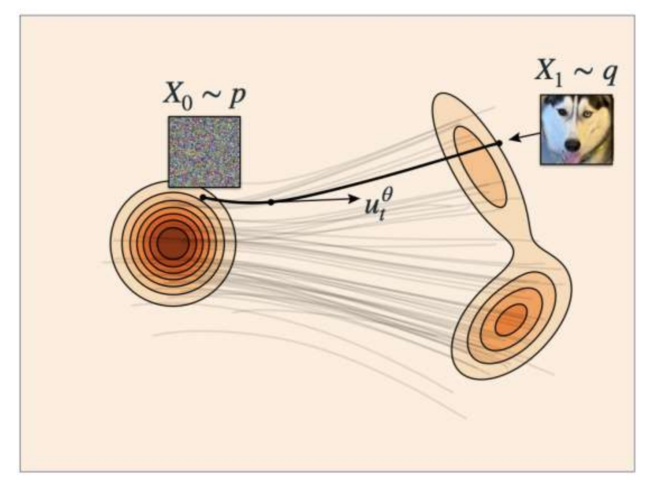


Figure 10: Flowing matching uses **Differential Equations** (ODE) to construct a path from Gaussian noise to target distribution.

During sampling, use a Differential Equation solver like Euler's method where the velocity (vector field) is a neural network. (Lipman et.al, 2024)

Results

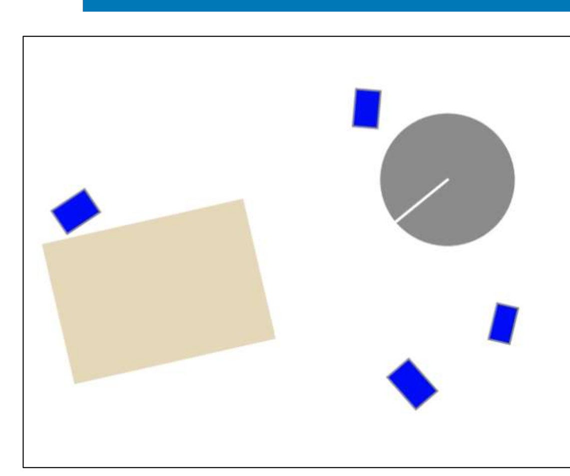
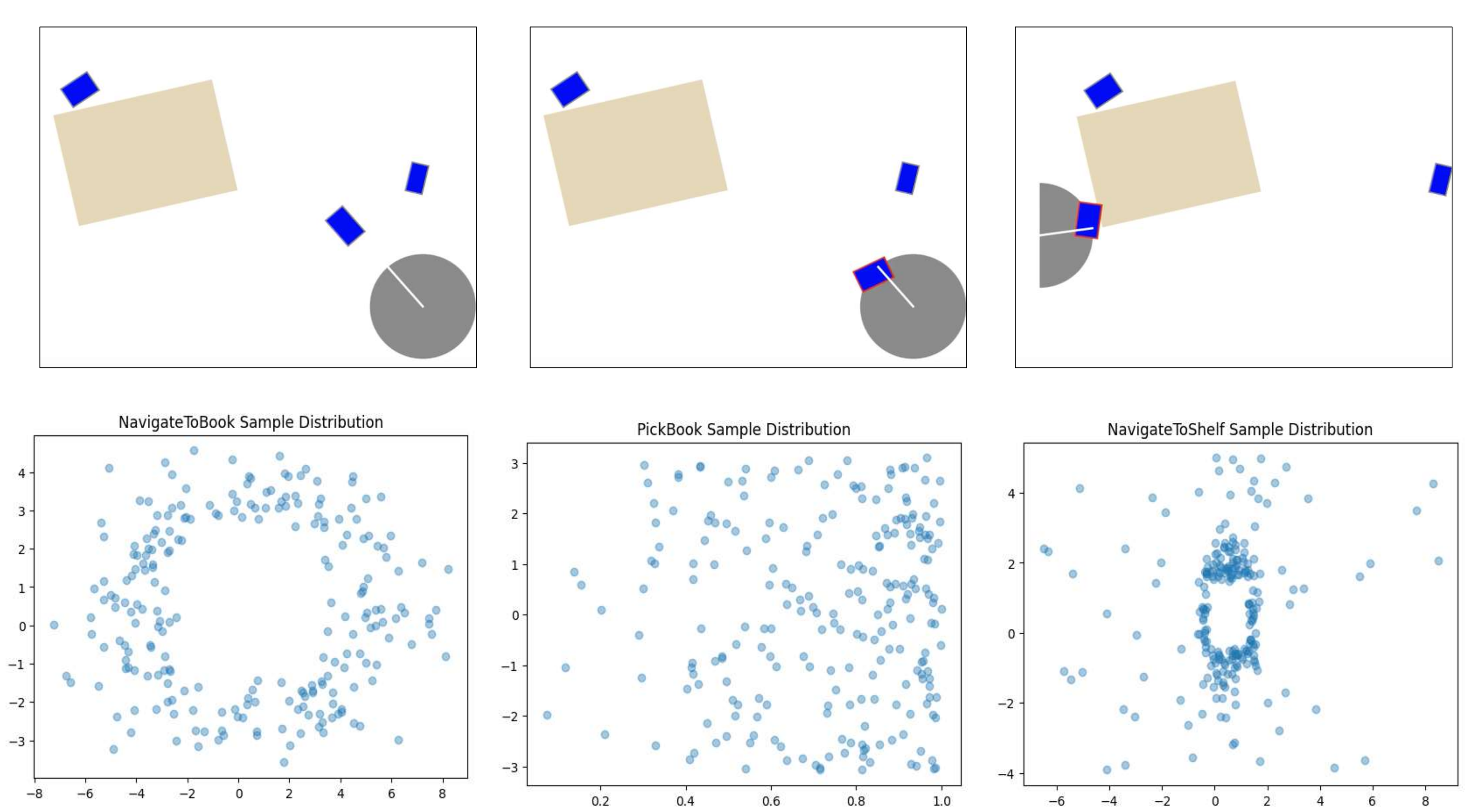


Figure 4: The planner tells robot the steps to complete placing books on the shelf. For each step, there is a distribution of successful samples for the generative model to learn.



Results

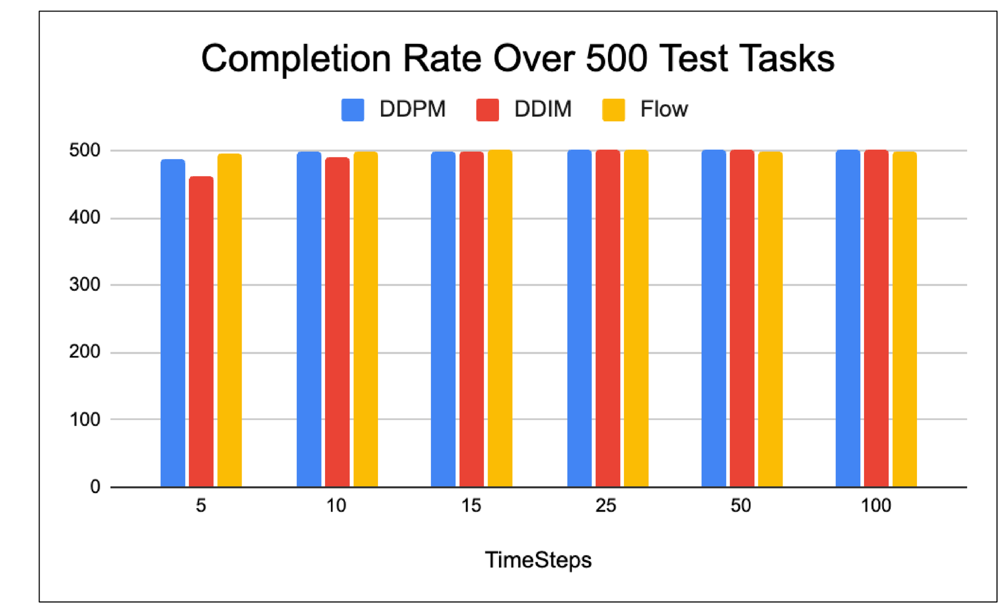


Figure 5: Success rate of the model at completing 500 test tasks (vary book locations, table length, etc.).

X-axis shows the sampling timesteps of the generative model.

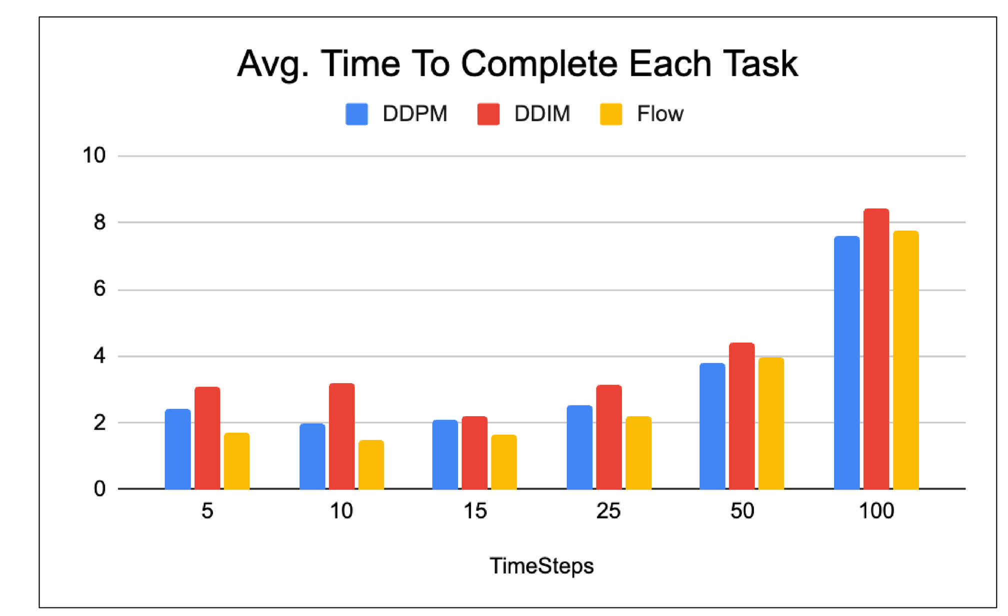


Figure 6: Time it took on average to compete 50 test tasks including planning, sampling and backtracking measured in **seconds**.

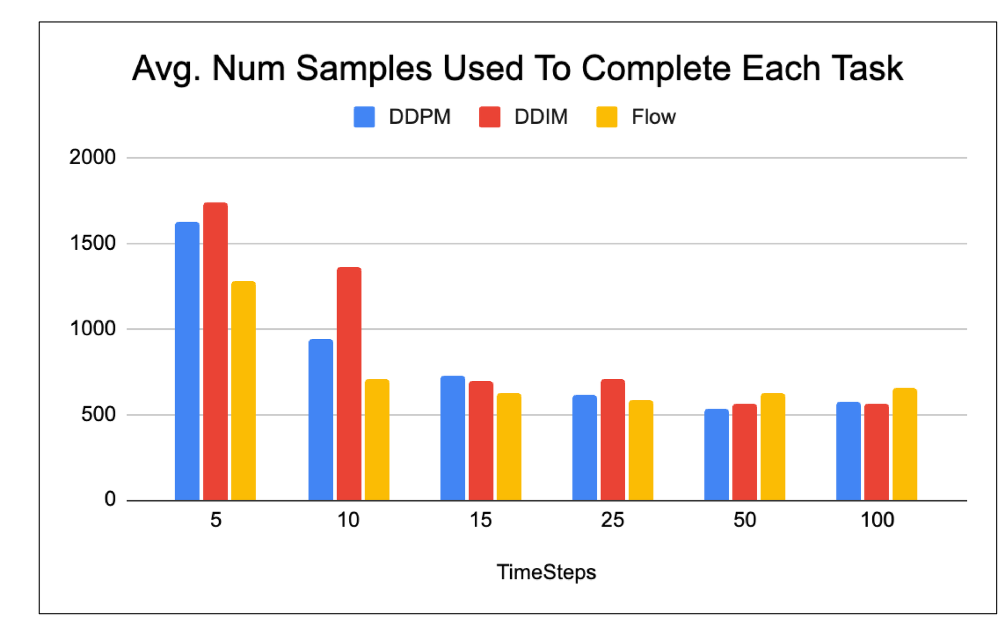


Figure 7: Number of samples generated for each task. More samples meant more times the sampler had to resample/backtrack, which led to a longer sampling time.

Conclusion

- Using Flow Matching reduced the number of action parameters at lower number of timesteps while maintaining the same level of performance.
- Reducing sampling steps worsened sampling quality, which led to more backtracking and samples generated.
- Planning speed improved only when the increase in sample was moderate.
- Learned samplers did not need to match demonstrations' distributions, as it can resample points faster to make up for the quality difference.
- Generative Models could sample with fewer number of timesteps for low-dimension data.

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